Chapter 7 — One-Time Events & Side Effects

> **Part I — Core Infrastructure** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 6 (UI State & UDF — `UiState`, `onEvent`, the `effects` `Channel`). This chapter deepens the **effect** half of UDF: how to deliver navigation, snackbars, dialogs, and toasts exactly once, lifecycle-safely. Reused by every feature and by Ch 9 (search retry), Ch 53 (navigation).
>
> **Reference surface in StreamCart:** the effect pipelines of Cart, Checkout, and Auth (navigate-on-success, show-error-snackbar).

7.1 Problem Definition and Real-World Context

In UDF, durable conditions are **state**; things that should happen **once** — navigate to checkout, show a snackbar, pop a dialog, fire a toast — are **effects**. Modeling effects wrong is one of the most common Android bugs:

- **Effect-as-state double-fire.** If "navigate to checkout" is a boolean in `UiState`, rotation re-reads the state and navigates *again*; a snackbar stored in state reappears every recomposition/rotation.
- **Lost effects.** If you emit an effect while the UI is backgrounded and the stream has no buffer, the snackbar/navigation is dropped — the user taps "pay," backgrounds briefly, and never sees the result screen.
- **Multiple-collector duplication.** With the wrong hot stream, two collectors (e.g., re-subscription on config change) each consume the effect, double-firing.
- **Leaky manual solutions.** The old `SingleLiveEvent` / "consume" boolean patterns are error-prone and don't compose with coroutines.

The correct model: a dedicated **effect stream** separate from state, delivered **exactly once** to a **single lifecycle-aware collector**. The main design decision is **`Channel` vs `SharedFlow`**, and getting it right (plus lifecycle-aware collection) is the whole chapter.

Three practical use cases

1. **E-commerce checkout (StreamCart).** "Payment succeeded → navigate to confirmation" must fire once, survive a brief backgrounding, and never re-navigate on rotation.
2. **Auth (Account).** "Invalid credentials → show snackbar" once per attempt, not again when the user rotates the login screen.
3. **OTT downloads (Watch).** "Download failed → show a retry dialog" delivered once even if emitted while the app is in the background.

Why this is architectural

Effect delivery semantics (exactly-once, lifecycle-aware, buffered) determine whether navigation and user feedback are correct across the Android lifecycle — a cross-cutting concern that every screen depends on.

7.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Deliver one-time effects (nav, snackbar, dialog, toast) separately from state.
F2	Each effect is handled **exactly once**.
F3	Effects emitted while the UI is not collecting are **not lost** (buffered).
F4	Collection is lifecycle-aware (paused when STOPPED, resumed later).
F5	Rotation/recomposition must not replay past effects.
F6	Support a single consumer (the screen) per effect stream.

Non-functional requirements

Attribute	Policy
Correctness	Exactly-once; no double navigation; no dropped feedback.
Lifecycle-safety	No work/delivery while destroyed; no leaks.
Simplicity	Prefer the simplest primitive that meets semantics.
Testability	Effects observable in tests (Turbine).

Scope

- **In scope:** `Channel`-based effects, `SharedFlow` alternatives, lifecycle-aware collection, buffering, exactly-once semantics, when to use each.
- **Out of scope:** navigation framework specifics (Ch 53), the state half of UDF (Ch 6), process-death (effects are transient and not persisted — Ch 8).

Assumptions & constraints

- Compose + coroutines; `lifecycle-runtime-compose`. Effects are **transient** — deliberately *not* persisted across process death (a pending navigation after process death is usually wrong; re-derive from restored state instead).

7.3 System Design Fundamentals

- **State vs effect.** State = "what the screen *is*" (re-rendered any time). Effect = "what should *happen* once" (consumed, not re-rendered).
- **`Channel` (rendezvous/buffered).** A hot, *consumed-once* stream. Each element goes to exactly one receiver; if no one is collecting, a buffered channel holds it until someone does. This is the natural fit for one-time effects with a single consumer.

- **`SharedFlow`.** A hot broadcast stream; `replay=0` means late collectors miss past emissions, and multiple collectors each receive every emission. Good for fan-out, risky for exactly-once (re-subscription can re-deliver recent items if `replay>0`, and multiple collectors duplicate).
- **Lifecycle-aware collection.** Collect effects inside `repeatOnLifecycle (STARTED)` / `flowWithLifecycle` so delivery pauses when the screen isn't visible and resumes on return — buffered effects are delivered then.
- **Exactly-once.** Achieved by (a) a single collector and (b) a consumed-once stream (`Channel.receiveAsFlow()`), or by a careful `SharedFlow` discipline.

Core insight: **a `Channel` exposed as `receiveAsFlow()`, collected by a single lifecycle-aware collector, gives buffered exactly-once delivery** — the correct default for UI effects. Reach for `SharedFlow` only when you genuinely need fan-out.

7.4 High-Level Architecture


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Components & ownership

- **ViewModel** owns a private `Channel<Effect>` and exposes it as a read-only `Flow` via `receiveAsFlow()`.
- **The Route composable** is the single collector; it collects lifecycle-aware and dispatches each effect to the right handler (NavController, snackbar host, dialog state).

Direction: effects flow ViewModel → UI, consumed once. Distinct from the state `StateFlow` (Ch 6), which is re-rendered.

7.5 Detailed Component Design


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Element	Responsibility	Semantics
`Channel<Effect>(BUFFERED)`	hold effects until consumed	exactly-once, single consumer, buffered
`receiveAsFlow()`	expose channel as `Flow`	each item to one collector
single lifecycle-aware collector	deliver on STARTED	no loss, no replay-on-rotation

Why `Channel.BUFFERED` (not rendezvous): a rendezvous channel (`send` suspends until a receiver exists) would stall the ViewModel coroutine if emitted while backgrounded, and risks dropping under cancellation. A buffered channel accepts the effect immediately and delivers when the collector resumes.

7.6 End-to-End Data Flow


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> The effect is emitted while backgrounded, **buffered**, and delivered exactly once on return — no loss, no double navigation.

7.7 Complete Kotlin Implementation

Package: `com.streamcart.core.ui.effect` and feature presentation packages.

7.7.1 Reusable effect-emitting base

```
abstract class EffectViewModel<E> : ViewModel() {
    private val _effects = Channel<E>(Channel.BUFFERED)
    val effects: Flow<E> = _effects.receiveAsFlow()

    protected fun emitEffect(effect: E) {
        // trySend is non-suspending; BUFFERED capacity accepts immediately.
        _effects.trySend(effect)
    }
}
```

7.7.2 A feature using it

```
sealed interface CheckoutEffect {
    data object NavigateToConfirmation : CheckoutEffect
    data class ShowSnackbar(val message: String) : CheckoutEffect
}

@HiltViewModel
class CheckoutViewModel @Inject constructor(
    private val placeOrder: PlaceOrderUseCase,
) : EffectViewModel<CheckoutEffect>() {

    // ... uiState: StateFlow<CheckoutUiState> as in Ch 6

    fun onEvent(event: CheckoutUiEvent) {
        when (event) {
            CheckoutUiEvent.Pay -> viewModelScope.launch {
                when (placeOrder()) {
                    is Outcome.Success -> emitEffect(CheckoutEffect.NavigateToConfirmation)
                    is Outcome.Failure -> emitEffect(CheckoutEffect.ShowSnackbar("Payment
failed"))
                }
            }
            else -> Unit
        }
    }
}
```

7.7.3 Lifecycle-aware single collection (Compose)

```
@Composable
fun CheckoutRoute(
    viewModel: CheckoutViewModel = hiltViewModel(),
    navController: NavController,
    snackbarHostState: SnackbarHostState,
) {
    val state by viewModel.uiState.collectAsStateWithLifecycle()
    val lifecycleOwner = LocalLifecycleOwner.current

    // Single, lifecycle-aware collector; delivers buffered effects on STARTED.
    LaunchedEffect(viewModel, lifecycleOwner) {
        lifecycleOwner.repeatOnLifecycle(Lifecycle.State.STARTED) {
            viewModel.effects.collect { effect ->
                when (effect) {
                    CheckoutEffect.NavigateToConfirmation ->
                        navController.navigate(Routes.Confirmation) {
                            popUpTo(Routes.Checkout) { inclusive = true }
                        }
                    is CheckoutEffect.ShowSnackbar ->
                        snackbarHostState.showSnackbar(effect.message)
                }
            }
        }
    }

    CheckoutScreen(state = state, onEvent = viewModel::onEvent)
}
```

7.7.4 The `SharedFlow` alternative (when fan-out is needed)

```
// Use when MULTIPLE independent collectors must each see the event
// (e.g., an app-wide event bus). replay = 0, extraBufferCapacity to avoid drops.
private val _events = MutableSharedFlow<AppEvent>(
    replay = 0,
    extraBufferCapacity = 16,
    onBufferOverflow = BufferOverflow.DROP_OLDEST,
)
val events: SharedFlow<AppEvent> = _events.asSharedFlow()
fun post(e: AppEvent) { _events.tryEmit(e) }
```

> With `SharedFlow(replay=0)`, a collector that re-subscribes (rotation) does **not** receive past events — good for no-replay, but **multiple** simultaneous collectors each receive every event (fan-out). For a single-consumer one-

> time UI effect, `Channel` is simpler and safer.

7.7.5 Module structure

```
:core:ui/effect  EffectViewModel
:feature:checkout/presentation  CheckoutEffect, CheckoutViewModel
:feature:checkout/ui            CheckoutRoute (single lifecycle-aware collector)
```

7.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Effect emitted while STOPPED	Lost effect	`Channel.BUFFERED` holds it; delivered on STARTED
Rotation re-subscribes collector	Replay/double-fire	`Channel` consumed-once; nothing replays
Two collectors (bug)	Each gets some effects (split)	Keep a single collector; `Channel` splits, doesn't duplicate
Rendezvous channel + backgrounded	ViewModel coroutine stalls	Use `BUFFERED`, `trySend`
Rapid effects	Overflow	Buffer capacity; `SharedFlow` overflow policy if used
`send` after scope cancel	Exception	`trySend` returns failure silently; or guard

- **`Channel` delivers each element to exactly one receiver.** With one collector that's exactly-once; with two collectors the effects are *split* between them (a reason to keep a single collector) — unlike `SharedFlow`, which *duplicates* to all collectors.
- **Lifecycle-aware collection via `repeatOnLifecycle`** cancels the collector in STOPPED and restarts in STARTED; the buffered channel replays nothing but delivers anything buffered meanwhile.

7.9 Failure Scenarios and Recovery

Failure	Detection	Response	UX
Effect emitted while backgrounded	lifecycle STOPPED	buffered, delivered on resume	feedback appears on return
Process death before delivery	process gone	effect lost by design; re-derive from restored state (Ch 8)	correct screen via state
Collector double-registered	two LaunchedEffects	split/duplicate effects	ensure one collector per stream
Effect storm (overflow)	buffer full	drop policy (SharedFlow) or bounded channel	newest kept
Navigation effect to invalid route	nav throws	validate route/args; backend authorizes (Ch 52)	safe fallback

 7.9 Failure Scenarios and Recovery

> **Deliberate non-persistence:** effects are *not* saved across process death. A navigation that was pending should be re-derived from restored state (e.g., "order placed" is state → the confirmation screen is reached from state, not a stale effect). Persisting effects causes wrong double-navigations.

7.10 Scaling the Implementation

- **Level 1.** A single `Channel<Effect>` per ViewModel + one lifecycle-aware collector.
- **Level 2.** A reusable `EffectViewModel<E>` base; sealed `Effect` per feature.
- **Level 3 — production.** Consistent effect handling across features; navigation effects integrated with the nav graph (Ch 53); snackbar host hoisted at the app scaffold.
- **Level 4 — large.** A shared effect contract + testing harness; typed effect handlers; analytics on effect delivery.
- **Level 5 — enterprise.** An app-wide event bus (`SharedFlow`) for cross-feature signals (e.g., "session expired → everyone logs out") alongside per-screen `Channel` effects, with clear rules on which to use.

Avoid overengineering: most screens need just a `Channel` and one collector. Don't build an event-bus framework for a single snackbar.

7.11 Alternative Approaches and Trade-offs

Approach	Semantics	Pros	Cons	Use when
`Channel` + `receiveAsFlow`	single-consumer, consumed-once, buffered	exactly-once, no loss, simple	one collector only	**default** for UI effects
`SharedFlow(replay=0)`	multi-consumer broadcast, no replay	fan-out	multiple collectors duplicate; late collectors miss; re-subscribe nuances	app-wide event bus / multiple listeners
`StateFlow` + consume flag	state-based	simple to reason	easy to double-fire; boilerplate to "consume"	avoid for effects
`SingleLiveEvent` (legacy)	single consumer	familiar in View code	fragile, not coroutine-native	legacy LiveData only
Effect in `UiState`	state	none for effects	re-fires on rotation	never for one-time effects

`Channel` vs `SharedFlow` (the core trade-off): `Channel` is *consumed-once* and point-to-point — ideal for a single screen reacting once. `SharedFlow` is *broadcast* — if two collectors exist they both react (duplication for effects, desired for a bus). For one-time UI effects with a single screen collector, use a `Channel`. For a genuine one-to-many signal, use `SharedFlow`.

7.12 Security Considerations

- **Don't carry secrets/PII in effects** (e.g., a full token in a "navigate" payload); pass ids and resolve securely (Ch 20); effects can be logged.
- **Navigation effects are not authorization** — validate deep links and privileged destinations server-side (Ch 52); a client effect can be triggered by a tampered client.
- **Redact effect contents in analytics/logs** (Ch 28); log the effect *type*.
- Client vs backend: an effect reflects a client decision; the backend must enforce the actual action (e.g., the order is authoritative server-side).

7.13 Performance and Resource Optimization

- **Lifecycle-aware collection** avoids doing effect work while the screen is invisible, saving CPU/battery and preventing crashes (navigating a destroyed controller).
- **Bounded buffers:** a small `Channel` buffer or `SharedFlow` overflow policy prevents unbounded memory under effect storms.
- **One collector:** multiple collectors waste work and cause duplication/splitting bugs.
- **Avoid heavy work in the collector:** dispatch to handlers; keep the collector thin.
- **Profiling:** watch for leaked collectors (a `LaunchedEffect` not keyed to the lifecycle), and for navigation crashes in Crashlytics indicating lifecycle misuse.

7.14 Testing Strategy

Effects are observable flows — test with Turbine; no UI needed.

```
@OptIn(ExperimentalCoroutinesApi::class)
class CheckoutEffectsTest {
    private val dispatcher = StandardTestDispatcher()
    @Before fun setUp() { Dispatchers.setMain(dispatcher) }
    @After fun tearDown() { Dispatchers.resetMain() }

    @Test fun `successful pay emits navigation exactly once`() = runTest(dispatcher) {
        val vm = CheckoutViewModel(FakePlaceOrder(success = true))
        vm.effects.test {
            vm.onEvent(CheckoutUiEvent.Pay)
            assertEquals(CheckoutEffect.NavigateToConfirmation, awaitItem())
            expectNoEvents()                // exactly once
        }
    }

    @Test fun `failed pay emits snackbar not navigation`() = runTest(dispatcher) {
        val vm = CheckoutViewModel(FakePlaceOrder(success = false))
        vm.effects.test {
            vm.onEvent(CheckoutUiEvent.Pay)
            assertTrue(awaitItem() is CheckoutEffect.ShowSnackbar)
        }
    }

    @Test fun `effect emitted before collection is buffered, not lost`() = runTest(dispatcher) {
        val vm = CheckoutViewModel(FakePlaceOrder(success = true))
        vm.onEvent(CheckoutUiEvent.Pay)      // emit before collecting
        vm.effects.test {
            assertEquals(CheckoutEffect.NavigateToConfirmation, awaitItem())  // buffered
        }
    }
}
```

Also: an instrumentation test that rotates the screen and asserts a snackbar is shown once (not twice). Fake the use cases; assert on the effect flow.

7.15 Production Observability

- **Metrics:** effect delivery count by type, navigation-effect failures, double-navigation incidents (should be 0), dropped-effect rate.
- **Logs:** `effect{type, screen}` (no payload PII), `nav{from,to}`.
- **Crash signals:** navigating a destroyed `NavController` (lifecycle misuse) → shows up as IllegalStateException; a reappearing snackbar on rotation → effect modeled as state.
- **Debugging:** "double navigation" → effect stored as state or two collectors; "snackbar never shows after backgrounding" → rendezvous channel or no buffer or collecting without lifecycle awareness.

7.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *State vs effect?* State = re-rendered condition; effect = one-time occurrence (nav/snackbar).
2. *Why not model navigation as state?* Re-fires on rotation (double navigation).
3. *What delivers exactly-once?* A single collector over a consumed-once stream (`Channel`).
4. *Why buffer effects?* So an effect emitted while backgrounded isn't lost.
5. *Why lifecycle-aware collection?* Pause when invisible; resume + deliver buffered on return; avoid crashes.

Channel vs SharedFlow

6. `Channel` *semantics?* Each element to exactly one receiver; buffered; consumed once.
7. `SharedFlow` *semantics?* Broadcast to all collectors; `replay` controls history.
8. *Which for a one-time UI effect?* `Channel` (single consumer, exactly-once).
9. *Which for an app-wide bus?* `SharedFlow` (fan-out).
10. *Two collectors on a* `Channel`*?* Effects are *split* between them (bug) — keep one collector.
11. *Two collectors on a* `SharedFlow`*?* Both receive every emission (fan-out).
12. `replay=1` *SharedFlow for effects — risk?* Re-subscription re-delivers the last effect (double-fire).

Kotlin/Compose

13. *How collect effects in Compose?* `LaunchedEffect` + `repeatOnLifecycle (STARTED)` (or `flowWithLifecycle`).
14. `trySend` *vs* `send`*?* `trySend` is non-suspending (use with BUFFERED); `send` suspends until capacity.
15. *Why* `receiveAsFlow()`*?* Expose the channel read-only as a `Flow`.

Lifecycle

16. *Effect emitted in STOPPED — what happens?* Buffered; delivered on STARTED.
17. *Rotation — do effects replay?* No (Channel consumed-once).
18. *Process death + pending effect?* Lost by design; re-derive from restored state.

Correctness

19. *How prevent double navigation?* Effect (not state) + single collector + consumed-once.
20. *Rendezvous channel pitfall?* `send` stalls if no collector; use BUFFERED.

Scalability

21. *Cross-feature "session expired" signal?* `SharedFlow` event bus.
22. *Standardize effects?* `EffectViewModel<E>` base + lifecycle-aware collector.

Security

23. *Secrets in effect payloads?* No; pass ids; effects may be logged.
24. *Is a nav effect authorization?* No; backend authorizes.

Testing

25. *How test effects?* Turbine on the effects flow; assert type + exactly-once + buffered.
26. *How test no-double-fire on rotation?* Instrumentation rotate test; assert single snackbar.

Production debugging

27. *Snackbar repeats on rotation — cause/fix?* Effect-as-state; move to Channel.
28. *Navigation lost after backgrounding — cause/fix?* No buffer / rendezvous / non-lifecycle collection.

Design-change challenges

1. **Add an app-wide "logged out everywhere" signal.** `SharedFlow` bus consumed by a top-level collector (Ch 22/42).
2. **Snackbar with an action (Undo).** Effect carries an action lambda id; handler wires `SnackbarResult`.
3. **Effect must survive process death (rare).** Don't — re-derive from persisted state (Ch 8).
4. **Many effects, risk of overflow.** Bounded buffer + drop policy; coalesce.
5. **Two screens listening to one ViewModel's effects.** Switch to `SharedFlow` or give each its own stream.

Mock walkthrough (abridged)

Clarify (one-time? single consumer? survive backgrounding?) → choose `Channel` + `receiveAsFlow` → single lifecycle-aware collector → show buffering guarantees → contrast with `SharedFlow` for fan-out → process-death non-persistence → testing.

7.17 Practical Exercises

1. **Beginner.** Add a `Channel<Effect>` to a ViewModel and collect it lifecycle-aware; fire a snackbar once. *Eval:* single delivery; no rotation replay.
2. **Intermediate.** Prove buffering: emit an effect before collection starts and assert it's delivered. *Eval:* not lost.
3. **Advanced.** Build an `EffectViewModel<E>` base and use it in two features; add navigation with `popUpTo`. *Eval:* reused; correct back stack.
4. **Production debugging.** Fix a screen that double-navigates on rotation (effect-as-state) and one that drops a snackbar after backgrounding (rendezvous/no buffer). *Eval:* exactly-once + no loss.
5. **System design challenge.** Add an app-wide `SharedFlow` "session expired" bus that logs out all screens, coexisting with per-screen `Channel` effects. *Eval:* fan-out works; no duplication of per-screen effects.

(Solutions withheld unless requested.)

7.18 Chapter Summary & Reference Sheet

Key decisions: effects are separate from state; default to a `Channel<Effect>` exposed via `receiveAsFlow()`, collected by a **single lifecycle-aware** collector → buffered exactly-once delivery; use `SharedFlow` only for genuine fan-out (event bus); **never** model one-time effects as state; **don't** persist effects across process death (re-derive from state).

Concepts: state vs effect, exactly-once, consumed-once vs broadcast, buffering, `repeatOnLifecycle`, Channel vs SharedFlow.

Common mistakes: effect-as-state (double-fire); `SharedFlow(replay≥1)` for effects; multiple collectors; rendezvous channel (lost/stalled); collecting without lifecycle awareness; persisting effects.

Checklist:

- ☐ effects separate from `UiState`
- ☐ `Channel.BUFFERED` + `receiveAsFlow`
- ☐ single collector
- ☐ `repeatOnLifecycle(STARTED)`
- ☐ `trySend`
- ☐ no secrets in payload
- ☐ not persisted
- ☐ `SharedFlow` only for fan-out.

Key APIs: `Channel`, `trySend` / `send`, `receiveAsFlow`, `MutableSharedFlow` / `asSharedFlow`, `BufferOverflow`, `repeatOnLifecycle`, `flowWithLifecycle`, `LaunchedEffect`, Turbine.

References:

- UI events (one-off events): https://developer.android.com/topic/architecture/ui-layer/events
- Channels: https://kotlinlang.org/docs/channels.html
- SharedFlow/StateFlow: https://developer.android.com/kotlin/flow/stateflow-and-sharedflow
- `repeatOnLifecycle`: https://developer.android.com/topic/libraries/architecture/coroutines#restart
- A safer way to collect flows: https://medium.com/androiddevelopers/a-safer-way-to-collect-flows-from-android-uis-23080b1f8bda

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; `EffectViewModel` + Channel + lifecycle collection + SharedFlow alternative; concurrency (split vs duplicate); non-persistence rationale; 5 levels; Channel-vs-SharedFlow trade-offs; security; perf; Turbine tests; observability; 28 Q&A + 5 challenges. ✅
- **Expand later:** navigation architecture (Ch 53), session-expired bus (Ch 22/ 42), snackbar-with-action patterns.
- **Interview coverage:** exceeds ≥25 bar.